

Software Requirements Specification (SRS)

Project Title: LokVeda — Your E-Gram Panchayat

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Version: v1.0.1

1. Introduction

1.1 Purpose

This document describes the Software Requirements Specification (SRS) for **LokVeda — Your E-Gram Panchayat**, an academic web-based application developed to digitally represent core functionalities of a village-level governance system.

The purpose of this document is to provide a clear understanding of the system's functionality, design constraints, and future scope for academic evaluation and documentation.

1.2 Scope

LokVeda is a web-based platform designed to simulate an **e-governance system** for Gram Panchayat operations.

The current implementation focuses on:

- Secure authentication
- Session management
- Role-based access (User, Staff, Admin)
- UI/UX interaction
- Location validation
- Modular frontend architecture

Advanced backend operations and real-world service processing are intentionally kept for **future scope**.

1.3 Definitions, Acronyms, and Abbreviations

- **OTP:** One Time Password
 - **UI/UX:** User Interface / User Experience
 - **SRS:** Software Requirements Specification
 - **Admin:** Panchayat authority
 - **Staff:** Panchayat staff member
 - **User:** Registered citizen
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1.4 References

- IEEE SRS Documentation Standard
 - Firebase Documentation
 - HTML5, CSS3, JavaScript (MDN Web Docs)
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2. Overall Description

2.1 Product Perspective

LokVeda is a **standalone web application** developed using frontend technologies and Firebase as a backend service simulator.

Firebase is treated as a **virtual backend**, where authentication, database access, and session validation are encapsulated within a single module (`firebase.js`).

2.2 Product Functions

The system provides the following high-level functions:

- Role-based login (User / Staff / Admin)
 - OTP-based authentication
 - Secure session creation and validation
 - Automatic session timeout handling
 - Location-based access verification
 - Greeting and personalization features
 - Modular navigation system
 - Dark mode support
 - Logout safety mechanisms
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2.3 User Classes and Characteristics

User Type	Description
User	Citizen registered in the village database
Staff	Panchayat staff with regional permissions
Admin	Panchayat administrator with national-level access

2.4 Operating Environment

- Web Browser (Chrome, Firefox, Edge)
 - Desktop and Mobile devices
 - Internet connection
 - Firebase backend services
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2.5 Design and Implementation Constraints

- Frontend-only implementation
 - Firebase used to mimic backend behavior
 - No real government database integration
 - Academic usage only
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2.6 Assumptions and Dependencies

- Users are pre-registered in the database
 - OTP delivery is assumed to be reliable
 - Browser supports modern JavaScript (ES Modules)
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3. System Features

3.1 Authentication System

- Aadhaar-based identity verification
 - OTP verification for secure login
 - Regex validation for input security
 - User-friendly status messages
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3.2 Session Management

- Secure session creation on login
 - Automatic logout on inactivity (15 minutes)
 - Session invalidation on improper logout
 - Login attempt tracking and cooldown mechanism
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3.3 Role-Based Access Control

- Separate dashboards for User, Staff, and Admin
 - Unauthorized access redirection
 - Session guard to prevent direct URL access
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3.4 Location Checker

- User access restricted to village boundaries
 - Staff access restricted to state boundaries
 - Admin access restricted within India
 - Border-level accuracy support
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3.5 Navigation and UI Module

- Dynamic navigation bar post-authentication
 - Conditional visibility of navigation elements
 - Search UI (functional design, logic pending)
 - Logout confirmation modal
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3.6 Greeting Bot

- Displays contextual greeting messages
 - Personalized based on user role and name
 - Time-based greeting logic
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4. External Interface Requirements

4.1 User Interface

- Clean and minimal UI
- Responsive design for different screen sizes
- Dark mode with user preference persistence
- Accessibility-friendly labels and prompts

4.2 Hardware Interfaces

Not applicable (Web-based application).

4.3 Software Interfaces

- Firebase Authentication
 - Firebase Firestore Database
 - Browser Local Storage
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4.4 Communication Interfaces

- HTTPS-based communication with Firebase services
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5. Non-Functional Requirements

5.1 Performance

- Fast page load time
 - Lightweight frontend modules
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5.2 Security

- OTP-based authentication
 - Session validation before page access
 - Prevention of unauthorized data access
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5.3 Reliability

- Session recovery handling
 - Controlled logout behavior
 - Error-handled authentication flows
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5.4 Usability

- Clear feedback messages
 - Minimal learning curve
 - User-friendly navigation
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5.5 Maintainability

- Modular JavaScript architecture
 - Encapsulated backend logic
 - Easy feature extension
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6. Future Scope

- Real backend implementation
 - Service application submission and tracking
 - Admin approval workflows
 - Document upload and download (PDF/DOCX)
 - Notification system
 - Analytics dashboard
 - Multi-language support
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7. Conclusion

LokVeda serves as a structured academic demonstration of an e-governance platform with a strong focus on authentication, session management, and modular frontend architecture. The system is scalable and designed for future expansion.
